

Safoura Banihashemi

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EDUCATION

M.Sc. Artificial Intelligence University of Bologna, Italy	Sept 2023 – 2026(Expected)
B.Sc. Mathematics and Applications Ferdowsi University of Mashhad, Iran	Sept 2018 – Aug 2022

PROJECT

Qwen3 Terminal GRPO Fine-Tuning Model: huggingface.co/safoura00/qwen3-terminal-grpo - Fine-tuned Qwen3-0.6B as a tool-use agentic RL system for Linux terminal command generation using GRPO (Group Relative Policy Optimization), implementing a custom RL training environment with a regex-based reward function driving agentic decision-making across 22 tasks. - Achieved 72.7% accuracy (16/22 tasks) after GRPO fine-tuning — a +59.1% improvement over the base model baseline (13.6%), with mean reward increasing from 0.136 to 0.841. - Trained on a single A100 GPU using 4-bit quantization (NF4) and LoRA adapters via HuggingFace TRL and PEFT.	Completed: May 2026
Bias Mitigation via Multi-Agent Collaboration (CAMEL-AI) - Implemented multi-agent collaboration using CAMEL-AI to reduce biased LLM responses without re-training. - Introduced a 'think tool' as a policy-improvement mechanism for agent reasoning, consistently outperforming the baseline multi-agent setup in ambiguous classification. - Achieved 99.07% accuracy on Disability status and 98.56% on Physical appearance classification.	Completed: Sep 2025
Legal Text Augmentation for Downstream Tasks (CAMEL-AI) - Developed legal text augmentation techniques using WordNet + GloVe embeddings and agent-based LLMs with CAMEL-AI to improve class balance in the Demosthenes dataset. - Addressed class imbalance by expanding underrepresented categories: Princ +25%, Aut +62%, Class +48%, Conc +100%.	Completed: Jul 2025
Brain Tumor Segmentation Built a U-Net CNN for brain tumor segmentation on the BraTS 2020 dataset (368 patients); implemented a custom Dice + Cross-entropy hybrid loss, achieving 76.1% Dice and 81.5% IoU.	Completed: May 2025
Large-Scale AI Research Graph Analysis Constructed a research graph using NetworkX over 7,700 AI papers (Hugging Face, 2023–2025); discovered heavy-tailed collaboration patterns and dominant hubs including Google and Tsinghua University.	Completed: Feb 2025
Sexism Detection — LLM Benchmarking Benchmarked Mistral and Phi-3.5 on the EDOS 2023 dataset via zero-shot and few-shot prompting; Mistral achieved 73.3% accuracy (vs. Phi-3.5's 54.1%), exposing critical gaps in implicit sexist language detection.	Completed: Jan 2025

SKILLS

Agents & LLMs: CAMEL-AI, LangGraph, LangChain, Quantization, HuggingFace TRL, PEFT, RL Fine-Tuning, Prompt Engineering

Frameworks: Tensorflow, Scikit-learn, NumPy, Pandas, PySpark, NetworkX

Languages: English (Proficient), Italian (Elementary)